Housing First or Treatment First?

Manuscript RED-D-26-00078

Letter to the Coordinating Editor

Summary of changes

Dear Professor Karabarbounis,

Thank you for the decision letter, and for leaving the door open to a resubmission in the form of an almost new paper. That is what this is. The framework has been rebuilt rather than patched, the calibration has been redone from primary data, and the validation section that the Associate Editor and Referee 1 both judged to be below the bar has been written from scratch.

This letter summarises the main changes. Four further letters respond to the Associate Editor and to each referee, comment by comment, with the responses in red.

Before the summary, one framing point. The single most useful thing in the reports was the shared diagnosis that the paper had too many moving parts and too little discipline on the parts that mattered. We have taken that literally. The model is smaller than the one you rejected in every dimension the Associate Editor questioned, and larger in exactly the three the Associate Editor asked us to develop: homelessness, substance use, and income. The parameter count did not grow. What grew is the share of parameters attached to a named source or an identified moment.

0. What has been built since this letter was drafted

This letter was first written when most of what follows was a plan. It is now built, and the summary below is what changed, why, and which comment each change answers. Nothing in it is a new promise.

- **The taste for drugs no longer changes when a household becomes addicted.** Referee 2 (C5), Referee 3 (M2a) and the Associate Editor (A8) all made the same objection: with the weight on drugs rising in d, reducing addiction and expanding housing become complements mechanically, before any behaviour responds. The weights are now fixed across the disorder in the code as well as on paper, and the Associate Editor's own suggested alternative is what carries the difference instead: only the Stone-Geary shifter varies. That also makes the model's central claim honest, since addiction is now transmitted across generations without any preference for drugs being transmitted.
- **The elasticity of drug use to housing access is now a parameter of the model rather than an implication of it.** This is Referee 2's C1, which Referee 3 and the Associate Editor (A13) named independently, and it was the sharpest comment in the reports. There is now a single parameter governing that elasticity, calibrated to the contrast between Milby et al.'s arms, and it is a *hazard* rather than a taste: recovery from a disorder is six times harder on the street than in a home. We first proposed putting it in the Stone-Geary shifter, making it depend on the housing chosen, and retired that. Housing here is a choice, so a preference that depends on it is a preference over a choice, and a user would gain a reason to choose the street because the drug is worth more there. The hazard also matches what the trial reports, which is abstinence by arm. One consequence is worth noting: flow utility is now the same

function of consumption, housing and drugs in *every* state and after *every* choice, so the first item above and this one are the same restriction rather than two.

- **Households now value their children at every age, not only at a terminal date.** A household that does not survive the year leaves a child whose productivity depends on whether the parent was addicted or homeless. Since mortality itself depends on drug use and on housing, the child now enters the drug and housing decisions rather than sitting at the end of life.
- **Four hazards vary with age that were constants.** Baseline survival is the SSA period life table rather than a fitted curve, which had understated mortality by nearly half at the top of the range. The homelessness and addiction mortality multipliers are derived jointly, since the homelessness ratio is measured on a population that mostly also has a disorder. And the onset of mental illness is backed out of the observed prevalence profile: a scalar onset produced prevalence *rising* with age where the data fall by a factor of eight.
- **Homeownership, mortgages and the housing market are out.** The Associate Editor's diagnosis was too many moving parts. This is the largest single cut: five subroutines, a mortgage contract, an endogenous house price, and half of the housing grid. The model is a renter model, which is what the questions need.
- **The drug price has a supply curve with an elasticity of two**, bracketed by 0.79 from the Washington marijuana reform and about 3.6 from the cigarette and gasoline literatures, rather than the flat curve of the previous version. Referee 3's M2(d).
- **Three numbers were wrong and are corrected**: the mortality multiplier for addiction was applied as an excess rather than a ratio, so the effective figure was 7.9 rather than the 6.9 we cite; the fiscal cost of a homeless household used a figure blended across the homeless and the precariously housed rather than the literally homeless one; and the homelessness target was a person rate where the model's agent is a household.
- **The disamenity of homelessness is priced from survey evidence rather than set to zero.** Referee 2 (C2) asked for it. It is now in the model, calibrated to Loubière et al.'s eight country contingent valuation, €23 a year on average to end homelessness, which is 0.08 per cent of average annual earnings, with the €57 among those willing to pay anything as the top of a reported range. It prices the headcount alone, with no extra weight on the homeless who carry a disorder, because that is what the survey asked about.
- **The stylized model no longer sets two parameters from one fact.** The Cobb-Douglas weight on housing was taken from the Consumer Expenditure Survey and the size of a rental unit from a separate housing cost to income ratio. Those are two external numbers aimed at one object, and nothing made the pair reproduce the survey's share once households choose among a discrete set of units, the street among them, with a voucher covering part of the rent for some of them: the model delivered sixteen per cent against the survey's 26.8. The weight now carries the preference and the unit size is calibrated to realised spending, which removes the double count.

One further change is presentational and worth flagging. The stylized model that opens the paper has been rebuilt with two periods rather than three, with the disorder realised at the transition so that mortality has something to destroy without a third period. It is explicitly qualitative: it reaches a drug spending share of about five per cent against eleven in the data, and we now say in the text why no parameter closes that gap in a model whose consequences arrive thirty years after the decision. The magnitudes are the quantitative model's job.

1. What was cut, what was added, and why

The Associate Editor asked whether six specific ingredients earn their place. We agreed on five of the six and cut them.

Ingredient	Verdict	Reason
Homeownership	Cut	The policy question is homeless against housed. Tenure added a large state and answered nothing. Also removes Referee 2's objection D1 about the housing grid, since the grid is now homeless plus three rental sizes and every point is pinned by a moment.
Unemployment insurance expiration	Cut	A separate benefit duration state that no moment identified.
Type dependent transfers and bequests	Cut	Replaced by a uniform lump sum. There was never an argument for the type dependence.
Housing market general equilibrium	Cut	You are right that the partial and general equilibrium long run outcomes were nearly identical. House prices and rents are now fixed.
Labour supply	Kept, coarsened	Referee 3 explicitly wants it, for the welfare cost of financing the policy with labour taxes. We keep it on a coarse grid and now report the decomposition that makes its role visible, so the two views can be adjudicated on evidence rather than on assertion.
Section 4.1.3 on intergenerational transmission	Rewritten	You were right that the language was decorative. There is now an actual intergenerational linkage, a warm glow over the child's newborn value, so the discussion has something to stand on. Referee 2's comment C3 asked for the same thing.

Against that, six things were added, all of them requested:

1. **Serious mental illness** as a persistent state that raises the propensity to use, lowers employment, and raises the risk of homelessness (Referee 1, comment 1a).
2. **Endogenous treatment.** Households with a substance use disorder choose whether to take up treatment at a utility cost, which raises the recovery hazard. Referee 1's comment 1b, that a policy called Treatment First contained no treatment, was the most damaging single sentence in the reports and it was correct.
3. **An equilibrium drug price.** The price of drugs now clears a supply curve rather than sitting at one (Referee 1 comment 1c, Referee 3 comment M2d, and your own request).

4. **Costly and imperfect enforcement** of the sobriety condition: a testing probability, a per test cost that enters the government budget, a detection probability below one, and a false positive rate (Referee 1 comments 4a, 4b, 4c).
5. **A direct productivity penalty** from substance use disorder, which is your comment A20. It is now the parameter that carries the income gradient among people with a disorder, and setting it to zero is a robustness run rather than a maintained assumption.
6. **A homelessness externality inside period utility**, which replaces the loss of consortium adjustment. This is developed in section 3 below.

2. The loss of consortium adjustment is gone

Referee 1 comments 3a, 3b and 3c, Referee 2 comment C2, Referee 3 comment M2c and your comment A9 all converge on the same object from different directions. Referee 1's version is the sharpest: an adjustment applied after the model is solved, which changes the policy ranking, and whose conservative value is arguably zero.

We accept this entirely. The parameter is deleted. In its place, Referee 2's suggested externality, a disutility that every household suffers from the size of the homeless population, enters period utility, where households internalise it in the same way they internalise everything else. Three things follow. The welfare ranking is now produced by the model rather than added to it. The externality is a calibrated object with a moment behind it rather than a number chosen after seeing the answer. And Referee 3's question M2c, where does the market failure come from if use is rational, now has an explicit answer inside the model rather than an appeal outside it.

We also follow your suggestion of reporting welfare and mortality as two separate dimensions rather than folding mortality into welfare through a single parameter.

3. Errors we found in our own previous version

Rebuilding the calibration from primary sources turned up six errors in the version you rejected. We list them here rather than letting the referees find them, and because two of them bear directly on whether the paper's central mechanism was ever properly disciplined.

1. **The homelessness target had a unit error.** The model's agent is a household. The target, 0.30 per cent, was built on a person count from the HUD point in time data. Converting the numerator to households, 467,020 homeless individuals plus 57,563 homeless family households, against 131,332,360 occupied units, gives 0.399 per cent. The model was targeting a homelessness rate about a quarter too low.
2. **The unemployment rate given a substance use disorder, 26.9 per cent, had no traceable provenance.** Measuring it directly in the 2024 NSDUH microdata for ages 26 and above, weighted, gives 16.25 per cent for moderate or severe drug use disorder. The old figure implied a relative risk of 5.49 against an overall rate of 4.9 per cent. The data give 2.44. The model's channel from substance use to employment was running roughly 2.25 times too strong.
3. **The mortality channel was running at roughly a fifth of its measured strength.** The homelessness mortality multiplier was a single flat 1.6. Baggett et al. (2013) give race

adjusted rate ratios of 8.6 for men and 9.6 for women at ages 25 to 44, 4.5 at 45 to 64, and 1.1 at 65 to 84. The multiplier is now age varying and built from that profile, and the life table has been replaced with a current vintage.

4. **Points 2 and 3 biased the paper in the same direction, and it is the direction that favours Referee 1's rival hypothesis.** Referee 1's comment 1d asks whether the mortality channel, rather than the labour market channel, is really what concentrates drug use at the bottom of the income distribution. With the employment channel more than twice too strong and the mortality channel at a fifth of strength, the previous calibration was stacked against our own claim. Referee 1 was asking the right question of a version of the model that could not have answered it honestly.
5. **The 37 per cent figure for substance use disorder among the homeless, which you flagged in comment A16, is the alcohol figure.** It is 36.7 per cent in Gutwinski et al. (2021). The drug use disorder figure from the same pooled evidence is 21.7 per cent. Your suspicion that the number looked high was exactly right, and the target has moved.
6. **The value of life constant was pinned by a moment that fixes its effect but not its level.** A structural check found that at the previous parameters, 64.6 per cent of the population had negative continuation value: they would rather be dead. For those households the mortality cost of drug use is a mortality benefit, which silently inverts the paper's central mechanism for precisely the population the policy targets. The constant is now pinned by the model implied value of a statistical life at 7.5 million dollars, which is what the paper always claimed to be doing.

We would rather report these than defend the previous numbers. Two of the six were found because a referee's comment forced us to look.

4. Calibration and validation

Referee 1's comment 2a, that Table 2 never reported the model's own moments, is the comment that cost the paper the most, because as he says it made the entire quantification impossible to assess. The new calibration table reports the target, the model value and the gap for all 24 targeted moments, and two further moments are held out entirely and reported as validation.

Every moment now carries a named source with a vintage, and a script in the replication package that reproduces it from the public file. Referee 1's comment 2e, that the price elasticity moment could not be replicated as described, is addressed by an appendix that gives the model counterpart of every moment explicitly.

The validation section is new. It contains three things:

1. Untargeted moments, including two housing cost burden moments that were demoted from targets once a better identification for the housing grid became available, so they are now genuine out of sample checks.
2. A mortality validation: the model's implied standardised mortality ratios for the homeless and for people with a substance use disorder, against Baggett et al. and Degenhardt et al.
3. A direct replication of Costantini (2025), described next.

5. The Costantini comparison, which the Associate Editor called a contradiction

The Associate Editor writes that Costantini finds Housing First cuts three year mortality by 4.6 percentage points while Treatment First has no long term health effects, "in contrast with the model simulations of this paper". We take this seriously, because if it were right it would be fatal. Having read his design carefully, we do not think it is, and we now show this rather than assert it.

Two premises in that sentence do not hold.

First, **neither of his coefficients is our Experiment 1**. Both are local average treatment effects against *no housing intervention*: he uses two instruments against a control arm that receives nothing. Both estimates therefore live on the generosity margin, assistance against none. Experiment 1 holds the number of assisted households fixed and varies only conditionality. His pair of coefficients does not identify the difference between two rationed regimes, which is why comparing them takes a separate test in his own paper rather than a subtraction of two rows.

Second, **his Treatment First is not our Treatment First**. His is the Veterans Affairs Grant and Per Diem programme: time limited transitional housing, a stay of six months to two years, after which the veteran must find their own housing. His own mechanism section attributes the null to that expiry. In his Figure 4, transitional housing recipients are *more* likely to be housed than voucher recipients in quarter one, because a bed is available immediately while a voucher must find a landlord, and then the effect decays to 9 percentage points by quarter twelve while the voucher arm peaks at 34. The programme does not fail to house people. It stops housing them. Our Treatment First is a permanent voucher whose *retention* is conditional on sobriety.

Rather than argue this, we ask the model both of his questions in his units and at his margin. Two arms, each with its own baseline and its own marginal expansion of access, each producing the Wald ratio that his instrument estimates:

Arm	Model	Costantini
Permanent unconditional voucher	**[TO FILL: HF LATE]**	-4.60 pp, 95% $[-7.93, -1.27]$
Sobriety conditioned voucher that expires at the Grant and Per Diem rate	**[TO FILL: TF LATE]**	near zero

Neither number is targeted. We will report whatever the model produces, including if it fails.

There is a further point in Costantini's favour that we had not expected. His cause of death decomposition finds the mortality effect running through exposure to the elements, assault, and blood borne infection from unsafe injection, and he states explicitly that Housing First has *no* effect on mental health care utilisation, ruling out mental health treatment as the channel. Those are exactly the channels this model has, and the one it deliberately does not. Independently, the At Home / Chez Soi trial finds housing moves mental illness symptom severity by nothing at any wave. Two very different designs agree, and together they support a modelling decision that would otherwise look like an omission.

6. Where we push back

Four places, all handled in detail in the individual letters.

1. **The Costantini comparison**, above.
2. **Abramson and Van Nieuwerburgh on the fiscal cost of homelessness** (comment A15). We have now read their paper. The 2,500 dollars per household per month is an *assumption* inside their calibration, stated as such, with the components listed in their footnote 11. It is not an independent estimate of the fiscal cost, so it cannot serve as external validation of ours. We report the comparison anyway, because the reader will want it, but we say what it is. We also run the sensitivity that both you and Referee 2 ask for.
3. **Referee 1's comparison of drug user income** (comment 2c i). His data figure of roughly 74 per cent is computed over people with a use *disorder*. The model figure he compares it with, 41 per cent, is over people who *use*. These are different populations. Measured consistently in NSDUH for ages 26 and above, the ratio is 0.951 for users and 0.687 for moderate or severe disorder. Both are now targeted, which is a sharper test than the one he proposed: use is close to income neutral, and it is *progression* to disorder that concentrates at the bottom. The underlying complaint, that the model made drug users too poor, was correct.
4. **Referee 3 and the Associate Editor disagree about labour supply.** Referee 3's minor comment 1 wants a clearer role for the labour supply elasticity in the welfare calculation. Comment A3 asks what labour supply is doing in the model at all. We keep it, coarsened, and report the decomposition, which is the only way to settle the question with evidence.

7. Presentation

Two experiments in the body, the rest in an appendix, as you asked in comment A18. Welfare and mortality reported as two dimensions. The word "addicted" replaced throughout with substance use disorder language, comment A19. The calibration section rewritten so that it reads as a set of identification arguments rather than a list of 22 targets, with parameters separated into those standard in the literature and those new to this paper, comment A17. A dynamic stylised model in the body, replacing the static appendix that Referee 1 said was claiming more than it established, comment A1 and Referee 1 comment 1e.

Referee 1 asks for proper transition paths rather than a several period partial equilibrium adjustment, comment 3d. With the housing market general equilibrium cut, a transition path iterates on the labour tax and the drug price alone, which makes it feasible. It is now solved properly.

Thank you again for the second look.

Sincerely,

Shahar Rotberg